Transfer: energy and work

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Learning target: Choose and justify energy methods in new physical situations.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 19.*
Both become 1/9 of their original values; both scale with v^2.

02. Worked example: explain the model
Gravitational loss 40 J splits into kinetic 30 and thermal 10 J.

03. Read and interpret the model
Gravitational U decreases, K increases; height vertical.

04. Apply the physics
40 J each.

05. Explain your reasoning
Thermal, sound, etc; not destroyed.

06. Review: connect an earlier idea *Review: Unit 4, Day 10.*
K=20 J.

07. Exit check
First power = 60/5 = 12 W; second = 60/6 = 10 W. The energy is unchanged; average power decreases to 5/6 of its original value.

08. STRETCH: test the model
Reduce drop to 1.5 m with 10 J loss, or keep 2 m drop with 20 J loss; other valid accounts accepted.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 19.*
Coupled energy becomes 3/5≈0.6 of the original; multiply the mass and fraction factors.

02. Worked example: explain the model
Gravitational loss 40 J splits into kinetic 30 and thermal 10 J.

03. Read and interpret the model
Gravitational U decreases, K increases; height vertical.

04. Apply the physics
Initial 60 J; bottom 16 J; loss 44 J;$\frac{1}{2}kx^2=16\rightarrow x=0.40$ m.

05. Explain your reasoning
Thermal, sound, etc; not destroyed.

06. Review: connect an earlier idea *Review: Unit 4, Day 10.*
$h=v_i^2/(2\ g)=20$ m.

07. Exit check
10 m/s; no atmosphere, ideal central gravity, no further thrust, ignores rotation/other bodies.

08. STRETCH: test the model
Initial K 6 plus gravitational 30 minus loss 11=25 J; $\frac{1}{2}kx^2=25$ gives$x=0.5$ m. At maximum compression speed zero.